

Which Hand-Motion Composition Axes Are Worth Measuring

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Abstract

Modular motion priors decompose hand motion into a bank of primitives, on the premise that novel recombinations remain usable, which is rarely measured. We build an instrument for measuring it: a paired split scoring a naive against an informed arm on held-out compositions, five gates that reject sweeps whose numbers cannot mean what they claim, and a control whose true penalty is zero. Over six axes of three public datasets, four carry difficulty. Every axis carrying it has positive interaction and none carries it without, though only one could have falsified that. A statistic on raw pose arrays finds that interaction in minutes, so a negative reading is a cheap reason to skip an axis. It licenses nothing in the other direction and ranks axes badly, its association with the penalty falling from $r = 0.843$ to $r = 0.463$. Where difficulty exists a modular bank pays a smaller penalty than a parameter-matched monolithic latent, on six of seven axes. The control bounds every headline, and at matched rate reproduces 90% of the normalized advantage, so we claim nothing there. On a Shadow Hand, sixteen of seventeen defined comparisons fail to separate novel from familiar compositions.

Index Terms—Dexterous manipulation, learning from demonstration, motion primitives, deep learning methods, evaluation and benchmarking.

I. INTRODUCTION

Motion priors for dexterous hands are usually monolithic, emitting one continuous latent per control step. LAMP [1] is the current instance for hands, and it rejects a quantized alternative because “codebook switches can turn small latent changes into abrupt hand-command jumps during contact-rich residual RL”. Whole-body animation went the other way. MotionBricks [2] builds a modular latent backbone over more than 350,000 clips, but a human composes its primitives through an authoring interface, so it does not test whether the units of a *learned* bank mean anything stable.

Both architectures assume that separately learned primitives recombine and that novel recombinations stay usable. Whether modularity buys that at the scale hand datasets offer is not obvious. MotionBricks trains on 10^5 clips; the largest public hand dataset used here provides 10^3 . Whether any of it reaches a robot is a further question, as a composition can be valid human motion and still unreachable on a hand whose fingers are spaced differently. We build a measurement for the first and report a null for the second; our contributions are as follows.

1. An instrument for measuring compositional difficulty, combining a paired split, five gates that reject sweeps whose numbers cannot mean what they claim (Section III-C, Table I), and a control whose true penalty is zero. Four gates exist because an earlier design returned plausible numbers it could not have earned.
2. The finding that compositional difficulty needs the two factors to interact, and that informative factors alone do not produce it, across six axes on three datasets. Only one of them could have falsified the necessity direction, so we state it as one-sided evidence. A statistic finds the interaction without training anything, so a negative reading is a cheap reason to skip an axis. It licenses nothing in the other direction, ranks axes badly, and the one we swept on its recommendation came back the hardest of the six (Section V-B).
3. A null on the hardware question, and an account of why it is a weak null. Three things weaken it: the retargeting it

rests on has no collision term, the banks behind it were trained on the one dataset our own instrument finds null, and the equivalence test that would license “the same” fails its own sensitivity check.

That transfer is our measuring apparatus rather than our contribution. Work carrying human hand-object motion onto real hardware [7], [13] is further along that axis than this paper, and the three modular architectures we compare are likewise prior art.

II. RELATED WORK

LAMP [1] learns a history-conditioned continuous latent prior over hand actions and uses it as the action space for real-world residual reinforcement learning. DexSynRefine [7] represents human hand-object motion with motion-manifold primitives and grounds it in physics with task-space residual RL. Neither evaluates whether a prior’s primitives generalize to combinations absent from training. Our parameter-matched baseline *is* a LAMP-style continuous latent, and our modular arm takes its backbone from MotionBricks [2], whose primitives a human authors through an interface; what we study is the learned bank that replaces them.

Modular motion models recur in mechanically distinct forms, from an authored-primitive latent backbone [2] to switching linear [3] and nonlinear [4] dynamical systems and mode-adaptive gating [5], all sharing the premise that separately learned primitives can be recombined. A survey of over 700 movement-primitive papers [8] lists composing primitives and evaluating frameworks among its open questions.

Recent work uses “compositional” in senses that do not answer that question. CoDex [9] composes the *stages* of a task and learns no motion prior, while UniCross [10] composes four hand-defined skills and object-centric neural fields [11] mix object-conditioned primitives, both evaluating on unseen *objects*. In each case, however, the held-out axis is the object or the task stage, and none holds out the pairing of two factors. Adversarial Concept Search [12] comes closer, predicting which concept combinations a model will fail on from the geometry of its learned representations, but it needs a trained

model and works on language, where the statistic of Section III-D is computed before any model exists. SPIDER [13] converts kinematic human demonstrations into dynamically feasible robot trajectories at scale; we claim no contribution to retargeting, and Section VI states why the transfer used here is weaker than the field’s standard.

III. METHOD

A. Hand representation

Motion is 27 anatomical degrees of freedom in native units: four per finger (MCP flexion and abduction, PIP, DIP), five for the thumb, six for the wrist. Normalization into $[-1, 1]$ comes from the joint-limit box, never from dataset statistics, which keeps a prior trained on one source meaningful on another.

Public hand datasets provide MANO [20] parameters, and projecting them onto an anatomical model requires three conventions: which Euler axis carries flexion, its sign, and which of the remaining two carries abduction. We infer all three per joint and assume none. We did assume them first, and the thumb and the fingers turned out not to follow each other’s convention.¹ Assuming costs in both directions, and we paid both. Take abduction as $(\text{flexion} + 1) \bmod 3$, correct for the fingers, and 100% of GRAB and DexYCB frames fall outside thumb_mcp_abd’s 20° range. Pool PIP variance across fingers instead, and their differing resting offsets count as flexion; on OakInk2 that selected the wrong axis and pinned index_pip_flex at its limit in every frame. Inferring per joint halves that dataset’s projection residual and takes its PIP joints from 37.3% inside their anatomical range to 99.9%. Fig. 1 shows the three datasets agreeing on a reference pose once the conventions are inferred.

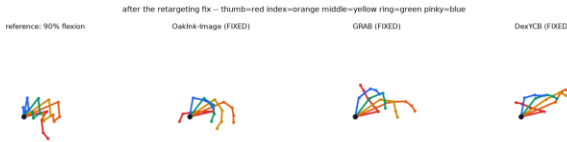


Fig. 1. The same 90%-flexion reference pose projected onto the anatomical model from each dataset, after the conventions are inferred per joint; thumb red, index orange, middle yellow, ring green, little blue. Before the fix an assumed abduction axis put 100% of GRAB and DexYCB frames outside thumb_mcp_abd’s range.

Section S1 audits how many degrees of freedom this leaves usable. Its count exempts the three wrist-rotation channels, which are constant zero in every MANO-derived bundle and would otherwise register as dead. MANO has no forearm, so its root rotation gives the hand’s absolute orientation in the capture frame rather than an articulation. Every error in this paper is averaged over 27 channels, three of them identically zero in both arms.

B. The two priors

Architecture. Both arms encode a 32-frame window of 27 anatomical DOF and reconstruct it under a bounded output, which makes joint-limit satisfaction architectural. The modular arm emits a relaxed categorical assignment over a bank of K primitives, one per frame, each a 32-dimensional

embedding, plus one 4-D style code for the window. That code is kept small so it cannot let the decoder bypass the bank. The assignment uses Gumbel-softmax [22] in training and a plain softmax at evaluation, so the reported numbers pass through a convex combination and never a hard switch. The baseline replaces the bank with a 12-D Gaussian latent, also per frame. Both encode with bidirectional two-layer GRUs of 256 units and decode with unidirectional ones, and their parameter counts agree to within 0.6% at the $K = 12$ that Section V-A’s sweeps use (Section S2).

Objective. Both are trained as beta-VAEs [23] on windowed reconstruction, with mean-squared error on pose, a first-difference term because the data is 30 fps motion, and β times a KL with a free-bits floor of 0.02 nats per dimension. The weight β is annealed over the first epochs, and validation is always scored at its final value. The modular arm adds four terms that shape the bank rather than the reconstruction: a switch penalty between adjacent frames, a per-frame entropy term, a usage-entropy reward and a load-balance term. A fifth is optional and off in Section V-A’s sweeps, and it penalizes the spread of the velocity profiles a primitive produces when decoded after several predecessors and from several initial poses, velocity rather than pose because pose is dominated by the initial condition. It optimizes the quantity Section V-C measures, which is why Section V-C reads its own number as a target being met.

Rate matching. The quantity to match is the rate of information flow, which the parameter count does not fix. A window-level latent would push 32 frames through one code while the bank emits an assignment every frame, so the baseline gets a per-frame latent too. On nominal capacity that leaves the baseline ahead, as a 12-D continuous Gaussian carries more per frame than a 12-way categorical whose 4-D style code is paid once per window. On realized rate, by contrast, the arms are not matched at all, and the gap favors the modular arm. The confirmatory axis gives 0.207 nats per frame for the baseline against 2.275 for the assignment channel, and two further sweeps give roughly elevenfold as well. Section S2 shows that the baseline’s KL penalty is nonetheless binding, that its latent is not collapsed, and that the assignment is nearly deterministic given the input, which is what makes it a high-rate channel.

The matched arm. Raising the baseline’s free-bits floor to 0.186 per dimension closes the gap. Its realized KL rises to 1.846 nats per frame against the modular arm’s 2.312, a ratio of 1.25, where the unmatched baseline in the same sweep sits at 0.206 for a ratio of 11.2. The confirmatory axis was re-swept on that basis, with all three arms trained together over the same 70 declared seeds, all of which returned every arm. That intervention produces a result, so Section V-A reports it, with the sensitivity check belonging to the intervention itself in Section S4. One caution applies to any comparison drawn across sweeps, and Section VI gives it.

C. Paired compositional measurement

We score the same target trajectories under two priors that differ only in whether their training data contained the target’s composition,

$$\text{penalty} = \text{MSE_naive}(T) - \text{MSE_informed}(T)$$

$$\text{paired} = \text{penalty_baseline} - \text{penalty_modular}$$

Both arms see the same number of training trajectories and are scored on identical test data, so trajectory difficulty and set size difference out and the arms’ evaluation noise is correlated. The unpaired alternative did not survive contact with the data as it then stood. The unseen set’s variance differed from the seen set’s by -5.9% to $+8.7\%$ by split seed, a pre-fix figure the release does not reproduce. On the corrected bundles the imbalance is -6.1% to $+10.7\%$, mean $+1.4\%$, and the gate passes on that mean. It passes on a statistic that is not the concern, though. The corrected per-seed spread is wider than the pre-fix one, so the risk the unpaired design runs is present in the release, and pairing removes it by construction.

TABLE I: the five gates, and what each caught

Gate	What it caught
primitive collapse	primitives <i>used</i> = 1.0, a modular model that is monolithic in disguise
informed coverage	held compositions absent from the informed set, which pins the penalty at zero by construction
composition leak	targets whose exact fine composition sat in the informed training set, against none for naive
DOF health	thumb <i>mcp</i> <i>abd</i> pinned at its limit in every GRAB and DexYCB frame under an assumed abduction axis
split balance	guard only; the risk that an unpaired design measures set difficulty rather than compositional difficulty

The fifth gate guards a risk the corrected data does not show. The other four each caught a defect in an earlier design, and two still exit 1 on the released artifacts, informed coverage on the confirmatory and affordance axes and DOF health at 40% and 33% against its own 25% threshold. Section S3 gives the transcript each gate was read from, the sweeps each was run over, and the breadth rule Section V-B applies. The DOF gate tests per-DOF pinned fraction and variance directly. It replaces a clip-fraction statistic, and the reason matters. That statistic is defined on values *before* clamping, so an adapter that passes it post-clamp values returns ≈ 0 whatever the data does.

D. Predicting which axes are worth measuring

A full sweep costs hours of GPU per axis, so a cheap statistic decides beforehand which axes repay that cost, and most axes do not. The compositional penalty asks whether knowing the *pairing* helps beyond knowing each factor separately, which is an *interaction*, so we fit two nested least-squares models per axis on PCA-reduced windowed poses,

$$\begin{aligned} \text{additive} & \quad \text{pose} \sim \text{left} + \text{right} \\ \text{full} & \quad \text{pose} \sim \text{left} + \text{right} + \text{cell} \\ \text{interaction} & \quad R^2(\text{full}) - R^2(\text{additive}) \end{aligned}$$

with a permutation null that shuffles labels over whole *trajectories*, since 30 fps windows within one trajectory are near-duplicates. A variance-explained statistic on the composition label alone does *not* work. Of the three real datasets scored on η^2 , GRAB’s excess ($+0.0538$) and OakInk-Image’s ($+0.0547$) are indistinguishable, yet GRAB’s swept

axis returns no compositional difficulty ($p = 0.494$) where OakInk-Image’s returns $+0.00982$. That row is scored on the id-prefix axis Section IV admits only for comparison, which is the axis whose η^2 was computed; it carries difficulty at $p = 1.1\text{e-}20$, so it serves the argument. The interaction term separates the two, at $z = 8.3$ on the confirmatory axis against 3.9 on GRAB’s.

E. Retargeting to the Shadow Hand

Robot model. The robot is the Shadow Hand E3M5 model from MuJoCo Menagerie [18], in MuJoCo [17]. Anatomical DOF map to its joints by role, corroborated by an independent check. The two models’ ranges were designed separately, so their agreement is evidence that the roles line up.² We transfer angles directly and clip them; rescaling between the limit boxes would hide the saturation this is meant to measure.³

Contact and penetration. Fingers resting together in a closed grasp register contacts and are valid poses, so the report separates *touching* from *penetrating beyond a tolerance* and counts only the second. At zero configuration the model has zero contacts and every geometry margin is 0, so measured penetration reflects real geometry, although that check is not in the release.

Abduction correction. We run coordinate descent on the abduction joints only, as flexion carries what the motion *is* and abduction carries how far apart the fingers are held, which is what the index-against-middle failure needs more of. Each step tries every abduction joint at $\pm 1.5^\circ$ and keeps the move that most reduces the deepest penetration, and which sign spreads which finger is read from the model.⁴

IV. EXPERIMENTAL SETUP

We use four public datasets (OakInk-Image [14], OakInk2 [15], DexYCB [16], GRAB [6]) plus two synthetic controls, one with planted via-point compositional difficulty and one purely interpolable. GRAB uses 8 of its 10 subject archives, which is how many downloaded. Sweeps run at a budget of 256 training trajectories per arm for 120 epochs, scored by a paired per-seed *t*-test with a sign test alongside, since a *t*-test alone cannot rule out a few extreme seeds carrying an effect. A result is declared only when both clear $\alpha = 0.05$, which is what “clears both declared tests” and “reaches neither” refer to below. The banks of Sections V-C and V-D use 200 epochs at stride 8 on full data.

The interpolable control returns no compositional advantage. At twenty seeds it gives $+0.00154$ with 12 of 20 ($p = 0.154$) at budget 256 and $+0.00383$ with 10 of 20 ($p = 0.438$) at budget 64 against the unmatched baseline. Against the rate-matched baseline of Section III-B the figures are $+0.00155$ with 13 of 20 ($p = 0.090$) and $+0.00324$ with 11 of 20 ($p = 0.49$), taking the two budgets in that same order.

We take composition axes from each dataset’s *own* taxonomy where one exists. OakInk ships object categories, functional classes and affordance attributes in its metadata, and all 100 object ids resolve. An earlier axis used the first character of the object id, which records which sub-collection an object came from, so results on it are carried through the paper for comparison alone.

V. RESULTS

A. The modular arm’s advantage, and what rate matching does to it

The advantage is how much smaller the modular arm’s compositional penalty is than the baseline’s, and how often. As Table II shows, it is positive on six of the seven interpretable axes, the six real ones and the synthetic control. It clears Section IV’s declared pair of tests on four; a fifth, the class axis, clears the t -test alone. The two exceptions are the axes whose difficulty is itself indistinguishable from zero: GRAB’s shape axis, positive but at $p = 0.269$, and OakInk2’s, the one negative value in the table. The difficulty the advantage acts on is a separate quantity, and we report the advantage first as the smaller and less certain of the two. The last column reads as the size of the problem and the second as how much of it the modular arm removes, so on the confirmatory axis $+0.00397$ of $+0.01042$ is 38%, and DexYCB’s sweep is omitted for the reason Section V-B gives.

TABLE II: the modular advantage, and the difficulty it acts on

data / axis	advantage (wins)	d	p	difficulty
synthetic, via-point	+0.00491 (17/18)	1.13	0.0002	+0.00904
OakInk-Image, category $\times$ intent	+0.00397 (48/69)	0.41	0.0012	+0.01042
OakInk-Image, class $\times$ intent	+0.00873 (9/12)	0.65	0.046	+0.01650
OakInk-Image, affordance $\times$ intent (exploratory)	+0.00365 (86/130)	0.33	0.0002	+0.00873
OakInk-Image, id-prefix $\times$ intent (deprecated)	+0.01022 (60/70)	0.91	<0.0001	+0.00982
GRAB, shape $\times$ fine intent	+0.00305 (25/40)	0.18	0.269	+0.00041
OakInk2, annotated transitions	-0.00010 (4/12)	-0.05	0.860	+0.00082

On that confirmatory axis the modular arm wins 48 splits in 69, seven in ten, which is the number to weigh before the effect size. It recurs on the class axis at $d = 0.65$, swept after the screen mis-ranked it (Section V-B), where the sign test is 9 of 12 at $p = 0.146$ and falls short of Section IV’s bar. We report that sign test because Section VI treats the same statistic as disqualifying on the affordance axis.

Scale is the first thing to rule out. The baseline’s absolute error is about a third higher than the modular arm’s, so a larger absolute penalty could be an artifact of error scale. Normalizing each seed’s penalty by its own naive error removes that confound, and the advantage then survives on two of the three official axes: $+0.076$ with 8 of 12 on the class axis, clearing neither test ($t p = 0.118$); $+0.045$ with 77 of 130 on the affordance axis, clearing both ($t p = 0.004$, sign $p = 0.043$); and $+0.046$ with 39 of 69 on the confirmatory category axis, which clears the t -test ($p = 0.021$) and not the sign test ($p = 0.336$). Normalizing therefore costs the confirmatory axis its

win rate, from 48 of 69 raw to 39 of 69. The normalized metric is a robustness check that failed. Two axes with no measurable difficulty return more on it than the confirmatory axis does, GRAB’s shape axis at $+0.116$ and the withdrawn DexYCB axis at $+0.043$ against the confirmatory $+0.046$. Wherever the modular arm reconstructs worse, dividing by naive error manufactures an advantage, and Section S4 gives that audit. The category axis carries the confirmatory claim and the affordance axis is exploratory, for reasons Section VI gives.

Rate matching. That normalization also decides how Section III-B’s rate-matched comparison reads, and on the two metrics it reads differently. Take the scale-normalized penalty first, whose premise is most clearly met here: the rate-matched baseline reconstructs better than either other arm. On that metric, matching the arms on realized information rate appears to remove nothing. The advantage is $+0.104$ against the unmatched baseline and $+0.111$ against the matched one, 52 of 70 at $d = 0.54$, clearing both declared tests. That reading does not survive the negative control, however. On *pc_easy*, the interpolable control, whose paired advantage is zero by construction, the same normalized contrast returns $+0.099$ at 18 of 20 and clears both tests, 90% of the figure just quoted. Normalization divides by each arm’s naive error, and on the control the matched baseline reconstructs 4.04 times better than the modular arm against 1.41 on real data, so dividing inverts the error-scale confound instead of removing it. On the raw penalty the same matching removes most of the effect, from $+0.00693$ at 50 of 70 wins and $d = 0.61$ to $+0.00254$ at 41 of 70 and $d = 0.23$, reaching neither declared test. The raw floor on the same control and arm is $+0.00155$, which is 61% of it. Both metrics therefore leave the comparison inconclusive, one because the effect is gone and the other because the instrument produces almost the same effect on data that has none, so we claim no modular advantage at matched rate.

Other decompositions. Two further decompositions, both taken off the shelf, beat the same parameter-matched baseline on synthetic data. *field* replaces the bank’s decoder with learned velocity fields under the same relaxed per-frame assignment, the switching-dynamical-system formulation of [3] and [4], and MANN blends expert weights per frame instead of committing to one [5]. The bank gives $+0.00491$ over 18 seeds ($t p = 0.0002$), *field* $+0.00567$ and MANN $+0.00968$ over 8 ($p = 0.001$ and 0.010), read against the baseline penalties Section S4 lists. Size does not explain those margins, as *field* and MANN carry 1,919,832 parameters against the bank’s 2,295,727 and only the bank is parameter-matched. All three carry Section III-B’s rate confound identically, and our claim is about the measurement, so the architectures being prior art leaves it standing.

B. Which axes carry compositional difficulty

The screened axes. We screened eleven axes, of which the six in Table III are those whose sweeps are interpretable; the other five all score negative and are listed with the release. One of them, DexYCB’s subject $\times$ shape, was also swept and is withdrawn below. Left and right give the two factors’ separate η^2 , excess is the interaction term of Section III-D

minus its permutation-null mean with the z beside it, and the last column repeats Table II’s difficulty.

TABLE III: the interaction screen against the measured penalty

dataset, axis	left	right	excess (z)	difficulty
OakInk-Image, affordance $\times$ intent	0.054	0.045	+0.0309 (14.9)	+0.00873
OakInk-Image, category $\times$ intent	0.097	0.045	+0.0238 (8.3)	+0.01042
OakInk-Image, id-prefix $\times$ intent (deprecated)	0.028	0.045	+0.0182 (12.9)	+0.00982
GRAB, shape $\times$ fine intent	0.028	0.044	+0.0063 (3.9)	+0.00041
OakInk-Image, class $\times$ intent	0.022	0.045	+0.0061 (4.1)	+0.01650
OakInk2, annotated transitions	0.188	0.243	-0.0120 (-2.6)	+0.00082

Informative factors are not enough on their own. OakInk2’s two factors are the most informative in Table III, at 0.188 and 0.243 against OakInk-Image’s 0.097 and 0.045, left factor first in each pair, and pairing them adds nothing measurable, while GRAB’s swept axis is less informative on both and also null. The axes carrying difficulty share the interaction term and the informative ones lack it. A practitioner choosing what to record should therefore measure the interaction. It is the harder property to arrange, and the one a factor-by-factor look at a dataset will miss.

Every axis carrying difficulty has positive interaction and none carries difficulty without it. Only one of the six could have shown otherwise, though. Falsifying it needs negative interaction together with measurable difficulty, and only

OakInk2’s excess is negative, at -0.0120 . The withdrawn DexYCB axis is negative too and would have been a second test had its sweep been interpretable, so the relation rests on one informative observation. Sufficiency is demonstrably false, since GRAB’s axis has interaction at $z = 3.9$ and returns $+0.00041$ at $p = 0.494$. Four of the six return a penalty distinguishable from zero, all with positive excess, three at $p < 1e-19$ and the fourth, at twelve seeds, at $p = 0.0018$. GRAB’s shape axis ($p = 0.494$) and OakInk2’s ($p = 0.115$) are the two that do not. Setting aside the deprecated id-prefix axis leaves five with three carrying difficulty, all from OakInk-Image, and the relation holds either way.

The screen does not predict magnitude. We swept six real axes before using it to choose one, and over the five that survive withdrawal the interaction excess tracks the baseline arm’s penalty at $r = 0.843$, already not significant at that sample size ($p = 0.073$). We then swept the class axis on the screen’s own recommendation, at twelve seeds, with the reading declared in advance: ranked fifth of six, it predicts a small penalty. It returned $+0.01650$, the largest difficulty we measured anywhere, and that one sweep took the association to $r = 0.463$ ($p = 0.355$). The collapse does not depend on the deprecated axis, as removing it leaves $r = 0.868$ over four ($p = 0.132$) before that sweep and $r = 0.446$ over five ($p = 0.451$) after. The screen ranks axes badly, and we found that out by trusting it. The obvious escape does not hold either. If the screen failed when the left factor is coarse, the id-prefix axis would fail too, and it has the same five groups and is ranked correctly. We therefore use the screen to decide which axis to try first, and let the sweep say how hard it is. Fig. 2 gives the same six axes as measured quantities, difficulty beside advantage.

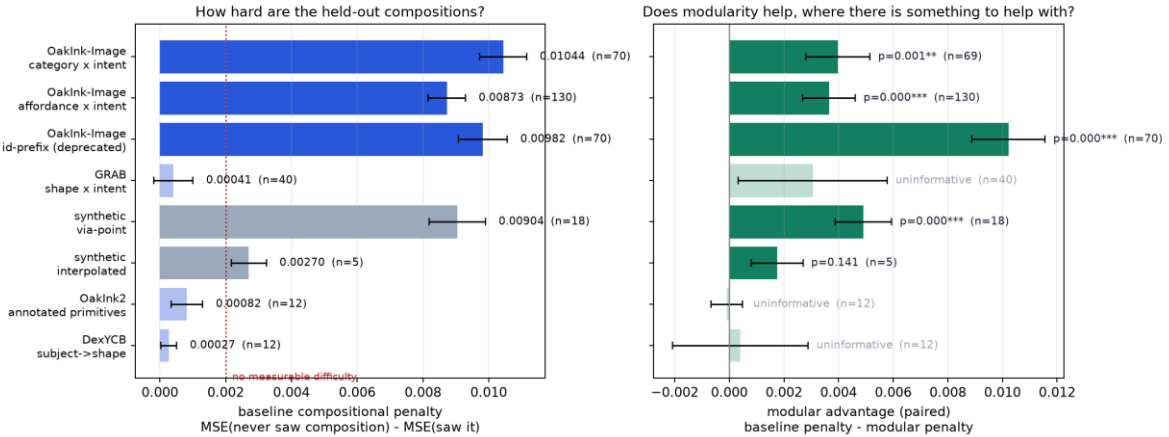


Fig. 2. Left, how much the held-out compositions cost a baseline that never saw them. Right, how much of that cost the modular arm removes, paired by seed. Faded rows are axes whose difficulty is itself indistinguishable from zero, where an architecture comparison is uninformative rather than negative. Every value is read from the released run artifacts.

A withdrawn axis. DexYCB’s sweep is excluded because it could not have measured what it claimed. It ran at $\text{min_chains} = 1$, which disables the restriction, as all sixty transitions in its bundle fall in a single fine group. The coverage gate then fails on all twelve seeds at both budgets,

the informed arm reaching a mean of 1.58 of the 4 held compositions. The failure is one of breadth, not depth, as the compositions it reaches carry 5.0 to 16.0 examples, inside the 3.8 to 37.3 band the other axes sit in. But a held composition the informed arm never sees is what Table I calls pinning the

penalty at zero by construction, and no configuration recovers it. DexYCB’s null belongs to the split, and the dataset is untested here. A second sweep is withheld for the same reason, and Section S4 gives it.

Coverage. We report the remaining axes case by case. Our own gate transcripts for two of them covered ten seeds of sweeps that used forty and twelve, so the claim they supported stood for longer than the evidence did. GRAB’s shape axis and OakInk2’s pass every row of their full sweeps, 80 rows and 24 respectively, which matters because they carry the nulls the necessity argument rests on. Thinness costs a penalty its size, so the class axis is measured conservatively, failing 6 of 24 rows on breadth and still returning +0.01650. The other three fail a minority of rows whose informed arms still reach 4 of 5 held compositions at 8.0 to 28.0 examples each, against DexYCB’s 1 or 2 of 4. Excluding every seed flagged moves the confirmatory advantage from +0.00397 over 69 to +0.00419 over 66 ($t p = 0.0010$), and Section S3 gives the rule with the per-row tallies.

What produces it. We do not know which collection property produces the interaction, and two candidates fail. Neither more pose-relevant intents nor a crossed design accounts for it, and Section S4 refutes both on the datasets’ own numbers. At $n = 11$ the interval on a correlation is wide, so neither is refuted outright, but neither is what the data show. The axes returning a penalty all belong to one dataset we cannot distinguish from the others on any property we measured.

C. Primitive identity

Reconstruction error says nothing about whether the bank’s units are units. The identity ratio is between-primitive over within-primitive variance in what a primitive produces (Fig. 3), so above 1 means its output is determined more by which primitive it is than by what preceded it. Pooled over 13 runs carrying Section III-B’s consistency term it is 1.841 ± 0.692 , with 11 of 13 above 1 ($t = 4.38$, $p = 0.0009$; sign test $p = 0.0225$) and a range of 0.87 to 2.84. It also replicates on the dataset that carries the difficulty. Those 13 runs are trained on OakInk2, which Section V-A finds null, and four banks trained on OakInk-Image under an identical recipe give 1.848, all four above 1 ($t = 5.07$, $p = 0.015$) at an $n = 4$ where the sign test cannot clear 0.05 and is not quoted. Read that ratio as a target being met, since the consistency term optimizes exactly this quantity.

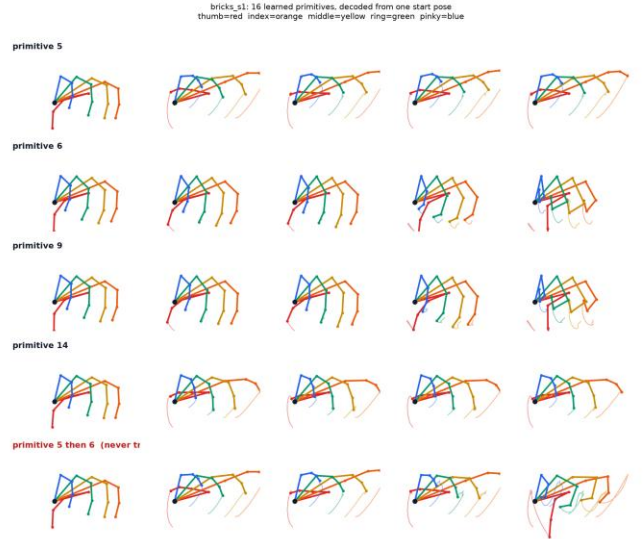


Fig. 3. Four of the sixteen primitives in one bank, each decoded from the same start pose and shown at five points through its window, with the trailing fingertip paths faded. The last row is one primitive followed by another, a pairing the bank never saw in training.

The ten banks trained without the term differ from the treatment arm in size or dataset, so we do not argue from them, and six of the ten are in fact the weight-zero arms of two smaller sweeps read against the wrong comparison. Those two sweeps hold size and dataset fixed at $K = 12$ and vary only the weight, and both move the same way, with mean identity rising from 0.677 to 1.773 in one and from 0.299 to 1.110 in the other, over three seeds each.

D. Transfer to the Shadow Hand

Setup. We attempted twenty-two comparisons between motion generated for unseen primitive pairs and for seen ones. A primitive pair is either seen or unseen and never both, so the pairing of Section III-C is unavailable here. Grasp retention is scored by Fisher exact, the rest by Welch’s unpaired t -test. Five are undefined because both arms are identically zero: three where the abduction or penetration measure never varied, and two grasp-retention cells at hard where neither arm retained anything (0 of 23 and 0 of 40). Fisher’s exact test returns 1.0 there where Welch’s returns an undefined statistic. Of the remaining seventeen, ten of them grasp-retention tests, four on abduction cost and three on penetration, sixteen fail to separate the arms. The seventeenth separates them in the opposite direction, at abduction cost $p = 0.037$ with the unseen arm worse, 3.32° for the seen arm against 4.64° for the unseen one. Four tests at $\alpha = 0.05$ give at least one such result about one time in five, which makes this consistent with no effect. It is still a contradiction, and we carry it forward as one.

Failure to reject is not evidence of similarity, so we test equivalence directly, and it does not survive its own sensitivity check. Pooled over five priors at 165 clips per arm, abduction cost is equivalent within 1° ($p = 0.008$) against a margin fixed in advance. But 60 of those clips per arm come from one prior whose correction never fired, all 120 of its values exactly 0.0. That block cannot fail a one-degree margin, and it collapses

the pooled variance; dropping it leaves 105 clips per arm at $p = 0.124$. Going back to the full 165 and dropping the prior that carries the reversal moves the difference from -0.36° to -0.06° , so two blocks carry the test in opposite directions. The test needs more clips than these banks can supply, and Section S5 gives the power calculation with the per-bank bounds. The other margins say little. Penetration never exceeds 2.3% after correction across these runs, retention within ten points is not established at easy ($p = 0.069$) or medium ($p = 0.074$), and the pass at hard is vacuous at 3.3% against 0.0%.

The retargeting baseline. Direct retargeting of the first 40 trajectories per dataset interpenetrates in 24.6%, 76.8% and 59.0% of frames on OakInk-Image, DexYCB and GRAB respectively, of which the abduction correction resolves 92%, 97% and 64% in that same dataset order. GRAB is the residual case, and the 2.3% ceiling quoted above for the bank-transfer runs does not describe it: the three datasets are left with 2.1%, 2.1% and 21.2% of frames penetrating, in the same order; Section S5 gives the clip rates. The failure concentrates by finger pair. On OakInk-Image, 1,617 of 3,960 penetrating contacts are index-distal against middle-distal, and all six of its most frequent body pairs are index against middle, as are DexYCB’s, where GRAB’s worst pair is little against ring. GRAB’s thumb and little-finger channels also arrive most clipped, at 10.2% of its DOF values, with THJ4 saturating in 87% of frames because the robot thumb cannot extend past zero where the human data goes negative. That is joint travel the correction cannot reach; what it addresses is finger geometry.

At the task level, four modular banks against four monolithic priors, sampled unconditionally on a MuJoCo grasp-retention task, retain 28.1% and 27.7% respectively at a pre-declared 480 clips per arm ($p = 0.94$), against 25.0% for uniform noise. Those monolithic priors carry 0.59 to 0.63 active units against the 0.996 to 1.000 of Section III-B’s baselines, so the null is between a healthy bank and a partly collapsed latent. An exploratory run on the same eight models at 160 clips per arm gave 35.6% for the modular arm against 26.9% for the monolithic one (Fisher $p = 0.117$). It appears here because the confirmatory script declared it would, that test existing precisely because the p value had been seen.

VI. LIMITATIONS

Section V-A’s two metrics disagree, and the negative control disposes of both. Matching takes the raw effect from +0.00693 to +0.00254, past neither test and down to that arm’s floor. On the normalized penalty it appears to remove nothing, yet the same contrast on `pc_easy` returns +0.099. Normalizing was meant to correct for a baseline that reconstructs worse, and the rate-matched baseline reconstructs better, so the correction runs the wrong way. No arm holds rate and reconstruction quality fixed at once, and no metric here separates the effect from the instrument. Intervening on the modular side is worse still. Charging the assignment its categorical KL zeroes that channel and raises reconstruction error by 57 to 72%, so its -0.00336 for `perframe_rate` against `modular_rate` (sign test $p = 0.041$, interpolable control) describes a model we broke.

Two sweeps of the same axis differ by more than either’s standard error, and this design separates neither cause. Training three arms consumes the random stream differently from training two, and the sweeps use `min_per_composition` 4 and 5, which changes the informed set and leaves the naive set identical. Read across sweeps, effect sizes carry that spread, whereas Section V-A’s rate-matched contrast sits within one sweep and is sound.

The paired penalty also returns a value where the truth is zero. `pc_easy` has no via-point offset, so its compositional penalty is zero by construction, yet in its own rate-matched bundle the baseline arm reports +0.00258 ($p = 0.0002$) at budget 256 and +0.00306 ($p < 0.0001$) at budget 64 over twenty seeds. Taking the budget-256 floor throughout, that is 25% of the unmatched baseline’s difficulty in the re-sweep (+0.01024, where Table II gives +0.01042 for the original) and 44% of the rate-matched baseline’s (+0.00586). Difficulty therefore reads partly off the measuring model, and on the arm this paper prefers the floor is nearly half of it. The same control bounds the advantage, and on one arm it bounds it away. Section IV’s unmatched arms differ by +0.00154 raw at budget 256, which is 39% of the confirmatory +0.00397, and by +0.018 normalized, again 39% of the normalized +0.046. Against the rate-matched baseline the raw floor is +0.00155, 61% of the whole effect, and the normalized floor is +0.099 at 18 of 20, which clears both declared tests. No headline should be read without these.

We truncated one sweep and extended another after seeing its p value. The category sweep declared 70 seeds, 69 of which returned both arms, and stopped at 52. The missing 17 are null alone (-0.00046 , $t p = 0.86$, 11 of 17 still favoring modular), so the first 52 were the favorable half and pooling moves d from 0.60 to 0.41. The affordance axis has the opposite defect. Seventy seeds gave +0.00270 at sign $p = 0.120$, sixty more were run, and the pooled +0.00365 passes at sign $p = 0.0003$. Having looked survives completing the sweep, so that axis is exploratory and the confirmatory claim stays with the category axis. The effect is small per split, too, as the modular arm wins 48 of 69 seeds on the category axis and 86 of 130 on the affordance axis, failing on roughly three splits in ten. The screen’s points are not independent either. Four of the six interpretable axes are OakInk-Image’s and share one intent factor on the same 770 trajectories, so six observations are closer to a one-dataset-against-two contrast than two or three sweeps elsewhere would settle.

Transfer is kinematic throughout: contacts are read off after `mj_forward` places the model, and nothing measures forces. Its baseline is weak in a specific way. Since DexPilot [19] the retargeting work we are aware of operates in task space, and the widely used dex-retargeting implementation of that objective, released with AnyTeleop [21], carries no collision term. The 24–77% interpenetration figure therefore characterizes that baseline and not the field’s standard, so we make no claim about pipelines carrying a task-space objective. We ran one, in which both arms penetrate in 100% of frames over four clips each, undefined for the reason Section V-D gives for its other zero-versus-zero cells. We report it because

it cuts our way and cannot support us. Those banks were trained on OakInk2, which Section V-A finds null.

Five further limitations are documented with the release and collected in Section S5, and two of them bound this paper’s claims directly. The advantage does not grow as data shrinks, over four sweeps at budgets 64 and 256 with two reversing and none significant. And contact does not explain the nulls, as GRAB’s 16 per-region channels leave the penalty at $+0.00032$ ($p = 0.813$).

The experiment that would settle this was not run. Sampling both priors unconditionally gives the architectures no channel through which to differ on grasp retention, which is why Section V-D’s task-level comparison cannot see the effect. Seeing it needs each prior used as the action space for a policy that is trying to grasp, as LAMP [1] and DexSynRefine [7] both do for continuous latents. LAMP’s discrete VQ-VAE arm finds gesture jumps at codebook switches, but it quantizes the commanded pose where our bank keeps a continuous decoder and makes only the selection discrete, so that artifact is not entailed. A bank of this form against a parameter-matched continuous latent is what we would run next.

VII. CONCLUSION

Before spending GPU on a compositional axis, run the interaction statistic of Section III-D over the raw pose arrays. It trains nothing and finishes in minutes, and a negative excess is a reason to look elsewhere, as none of the axes we measured carried difficulty without positive interaction. A positive one licenses less than it appears to, as GRAB’s shape axis clears the same bar and returns $+0.00041$ at $p = 0.494$. Where difficulty is present the modular bank pays the smaller penalty, but at matched rate we claim nothing for modular structure, since our own control manufactures most of the effect we would be claiming. A reader who wants that settled should look to a bank used as a policy’s action space, which is where the architectures were proposed and where we did not test them. We stop short of a dataset design rule. Every axis that returned a penalty belongs to one dataset, and across the eleven screened axes no collection property we measured predicts which.

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Footnotes

¹ MANO’s zero pose is a flat hand chosen for compatibility with SMPL [20], and its release carries a kinematic tree with no per-joint rest rotations, so every joint frame is parallel to the template’s. The thumb’s is therefore no better aligned to its own anatomy than a finger’s, and its flexion and abduction fall on different Euler components.

² index_mcp_abd [−20, 20] against FFJ4 [−20, 20], thumb_mcp_abd [−10, 10] against THJ3 [−12, 12], thumb_ip_flex [−15, 80] against THJ1 [−15, 90]. Where they agree less well (thumb_mcp_flex [−10, 55] against THJ2 [−40, 40]) the per-joint clip rate reports it.

³ The Menagerie model has 24 joints but 20 actuators, each finger’s two distal joints sharing a tendon, so writing qpos commands PIP and DIP independently where the physical hand cannot.

⁴ Shadow’s abduction axes are [0, −1, 0] for index and middle and [0, 1, 0] for ring and little, so a hand-written rule would be right for half the hand.